

Reading List — Reconstruction / Commensurability

Patrick S. Mahon
Personal reference

July 2026

The object. An observation protocol (windows on a symbolic dynamical system) is *commensurable* when pairwise-coherent window-statistics (EA) force realisability by a declared class $\text{PR}(\mathcal{R})$; geometrically iff the coherence polytope C equals the realisability polytope R . The paper is a *bridge across four fields*; the open capstone (universal impossibility) reduces to one lemma, **L-B**, living in the Perron–Frobenius + signed-graph corner. Paper skeleton: `papers/reconstruction/reconstruction_skeleton.tex`; proved body archived at `papers/reconstruction/notes/unsorted/joint2_wielandt_finding.md` and `notes/unsorted/parity_only`.

Diagnostic first (do this before reading anything). Skim two intros in one sitting: Abramsky–Brandenburger §§1–4 [2] (field 1) and Wainwright–Jordan §3–4 [12] (field 2). Whichever frame feels *alien* is the primary gap. Most readers missing “a whole field” here are missing polyhedral combinatorics (field 2) or Perron–Frobenius (field 4) — the contextuality and dynamics sides are more widely discussed. The two fields already owned, skip; go straight to the one that felt foreign.

Sources in dependency order, each with the question to read it for.

1 Contextuality / the sheaf picture (the FRAMING)

Supplies “coherent windows that come from no global process.” $C = R$ is their marginal polytope; EA = compatible marginals / no-signalling; $\text{PR}(\mathcal{R})$ = a global section in class $\mathcal{R}$; contextual = obstruction to a global section.

Abramsky–Brandenburger 2011 [2], *The Sheaf-Theoretic Structure of Non-Locality and Contextuality*. THE anchor. Read §§1–4.

- State “contextual = no global section of the presheaf of local sections.” Match it term-for-term to the paper’s $C \setminus R \neq \emptyset$. Which of their objects is the paper’s *context*, which its *protocol*?
- Where does *no-signalling* / *compatible marginals* enter? Confirm it is exactly EA (pairwise agreement on shared events).

Fine 1982 [6], *Hidden variables, joint probability, and the Bell inequalities*. The classical fact under everything.

- State “compatible marginals $\Leftrightarrow$ a global joint exists” for the cyclic case. This is $C = R$ specialised; read it as the sanity floor.

Abramsky 2013 [1], *Relational databases and Bell’s theorem*. The bridge tying field 1 to fields 2–3; unusually readable.

- Why is “acyclic reporting $\Rightarrow$ commensurable” *literally* the database running-intersection / Vorob’ev theorem? Identify the junction-tree glue. This is the paper’s trichotomy (acyclic arm).

2 Polyhedral combinatorics / marginal polytopes (the MACHINERY)

Makes “coherence forces a process” COMPUTABLE. If STAB/FSTAB, total unimodularity, fractional vertex, marginal polytope feel foreign, THIS is the gap.

Wainwright–Jordan 2008 [12], *Graphical Models, Exponential Families, and Variational Inference*, §3–4. The single best entry point. Their *marginal polytope* = the paper’s C .

- Define the marginal polytope as the convex hull of realisable margins. Map it to $R = \text{conv}\{\text{cell-incidence vectors of global configs}\}$. Which inequalities cut out C ?

Schrijver 2003 [11], *Combinatorial Optimization (or Theory of Linear and Integer Programming)*. TU, König, integral polytopes. Read the TU + stable-set chapters.

- State the sufficient condition: a $\{0, \pm 1\}$ matrix with each column signable so every row’s two nonzeros are opposite-sign is TU. This *is* taming (3), signability. Where does the frame’s bipartition supply the signing?
- Integral-polytope $\Leftrightarrow$ TU-with-integral-RHS: confirm this is the whole content of “ $C = R$ via integrality.”

Nemhauser–Trotter 1974 [9] + **Chvátal 1975** [4], the STABLE-SET polytope. STAB vs its fractional relaxation FSTAB, and when they coincide.

- State: $\text{FSTAB}(G) = \text{STAB}(G) \Leftrightarrow G$ bipartite (König). This is exactly the parity theorem’s even-case engine. Why does the odd cycle give a unique half-integral vertex $u \equiv \frac{1}{2}$? (That vertex is the PR-box.)

(context, not required) **De Loera–Onn 2004** [5] + **Průša–Werner 2015** [10], universality. Why every rational polytope is a coherence polytope $\Rightarrow$ no finite model-level taxonomy $\Rightarrow$ the classification must be posed at the PROTOCOL level.

3 Symbolic dynamics (the OBJECTS observed)

The language ρ , the shift of finite type, the transfer digraph, recurrence around a ring. If “golden-mean shift” / “SFT” is unfamiliar, this is the other likely gap.

Lind–Marcus 1995 [8], *An Introduction to Symbolic Dynamics and Coding*. THE textbook; only three chapters needed.

- Ch. 2 (shifts of finite type): define ρ = legal adjacent pairs; the golden-mean shift = “no 11.” The paper’s *language* is an edge-SFT.
- Ch. 4 (the transfer/adjacency matrix): the digraph on states with arc $a \rightarrow b$ iff $(a, b) \in \rho$. The paper’s *transfer digraph*; the *layered ring* is this SFT observed cyclically over $\mathbb{Z}_L$.
- The Perron–Frobenius chapter: primitivity/irreducibility of the transfer matrix — hands you straight into field 4.

4 Perron–Frobenius + signed graphs (the ENGINE of the open lemma L-B)

Where the FORCING LEMMA actually lives. “Primitive,” “period/imprimitivity,” “Wielandt index,” “cyclic classes” are all Perron–Frobenius; “balanced signed graph” is Zaslavsky. L-B’s two exclusions are PF statements about the DERIVED pair-graph. **This + the last source are the narrow path to L-B — not the whole stack.**

Brualdi–Ryser 1991 [3], *Combinatorial Matrix Theory*, Ch. 3. Primitivity, the Wielandt bound, imprimitivity/period, cyclic structure. The exact toolkit L-B needs.

- State: a primitive $n \times n$ 0-1 matrix has $M^k > 0$ for all $k \geq (n-1)^2 + 1$ (Wielandt). The length-availability half of “primitive $\Rightarrow$ cofinitely unsafe.”
- State the period/imprimitivity dictionary: strongly-connected digraph has period $d \Leftrightarrow$ vertices d -partition into cyclic classes arcs advance through $\Leftrightarrow$ fibered over $\mathbb{Z}_d$. This is taming (8)=grading. Primitive = period 1 = NO grading.
- **The two L-B exclusions, in PF language:** (i) primitive $\Rightarrow$ the *pair-graph* $\bar{D}$ is APERIODIC (has an odd-length closed walk); (ii) primitive+s.c. $\Rightarrow \bar{D}$ carries a CROSSED cycle (its σ -cover is connected). Read Ch. 3 asking: what forces aperiodicity/connectivity of a DERIVED graph from primitivity of the base? That is the open content.

Horn–Johnson [7], *Matrix Analysis*, Ch. 8 (alternative register). Cleaner analytic Perron–Frobenius if preferred to the combinatorial treatment. Same targets as Brualdi–Ryser Ch. 3; pick one, not both.

Zaslavsky 1982 [13], *Signed graphs + his matrices in the theory of signed graphs* survey. Balance theory.

- State: a signed graph is BALANCED $\Leftrightarrow$ 2-colourable $\Leftrightarrow$ no negative (odd-sign) cycle $\Leftrightarrow$ the sign is a coboundary. The paper’s “monodromy+1 balanced” IS this. Confirm L-A: safe-on-evens $\Leftrightarrow$ the pair-graph signed by (monodromy+1) is balanced. (The cycle-space $\mathbb{Z}_2 \times \mathbb{Z}_2$ argument is Zaslavsky’s balance read through two invariants at once.)

5 Notes

- **Start (if contextuality + dynamics already known):** Wainwright–Jordan §3–4 (marginal polytope = C) $\rightarrow$ Nemhauser–Trotter/Chvátal (STAB/FSTAB, König = the parity engine) $\rightarrow$ Brualdi–Ryser Ch. 3 + Zaslavsky (L-B’s home). That trio is the paper’s spine and the open lemma; the rest is framing pickable from Abramsky–Brandenburger §§1–4 on demand.
- **Start (if the whole thing is foreign):** Abramsky 2013 (relational databases / Bell) first — it ties contextuality to combinatorics in the most readable single paper here — then diagnose which of fields 2/4 is missing and go there.
- **The narrow path to the one open lemma:** Brualdi–Ryser Ch. 3 (primitivity / Wielandt / imprimitivity) + Zaslavsky (signed-graph balance). L-B is written entirely in those two vocabularies; the other three fields are not needed to attack it.
- Asymmetry vs the σ -essential thread: there the DST + set-theory were the gaps and the OML cluster was owned. Here the likely gaps are polyhedral combinatorics and/or Perron–Frobenius; contextuality/dynamics are more often already-held.

References

- [1] Samson Abramsky. Relational databases and Bell’s theorem. In *In Search of Elegance in the Theory and Practice of Computation*, volume 8000 of *Lecture Notes in Computer Science*, pages 13–35. Springer, 2013.
- [2] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13(11):113036, 2011. arXiv:1102.0264.
- [3] Richard A. Brualdi and Herbert J. Ryser. *Combinatorial Matrix Theory*. Cambridge University Press, 1991.

- [4] Vašek Chvátal. On certain polytopes associated with graphs. *Journal of Combinatorial Theory, Series B*, 18:138–154, 1975.
- [5] Jesús A. De Loera and Shmuel Onn. The complexity of three-way statistical tables. *SIAM Journal on Computing*, 33(4):819–836, 2004.
- [6] Arthur Fine. Hidden variables, joint probability, and the Bell inequalities. *Physical Review Letters*, 48(5):291–295, 1982.
- [7] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2nd edition, 2013.
- [8] Douglas Lind and Brian Marcus. *An Introduction to Symbolic Dynamics and Coding*. Cambridge University Press, 1995.
- [9] George L. Nemhauser and Leslie E. Trotter. Properties of vertex packing and independence system polyhedra. *Mathematical Programming*, 6:48–61, 1974.
- [10] Daniel Průša and Tomáš Werner. Universality of the local marginal polytope. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 37(4):898–904, 2015.
- [11] Alexander Schrijver. *Combinatorial Optimization: Polyhedra and Efficiency*. Springer, 2003.
- [12] Martin J. Wainwright and Michael I. Jordan. *Graphical Models, Exponential Families, and Variational Inference*, volume 1 of *Foundations and Trends in Machine Learning*. Now Publishers, 2008.
- [13] Thomas Zaslavsky. Signed graphs. *Discrete Applied Mathematics*, 4(1):47–74, 1982.